

X e Y indipendenti e binomiali cioè $X, Y \in \{0, 1\}$

$$P[X=1] = p$$

$$P[Y=1] = q$$

Trova la PDF e l'entropia del $\max(X, Y)$

$g(X, Y)$ con $g(x, y) = \max(x, y)$

$$X, Y \in \left\{ (0, 0), (0, 1), (1, 0), (1, 1) \right\}$$

$$\left\{ (1-p)(1-q), q(1-p), p(1-q), pq \right\}$$

$X \backslash Y$	0	1
0	$(1-p)(1-q)$	$(1-p)q$
1	$p(1-q)$	pq

$$g(X, Y) \in \{0, 1, 1, 1\} \equiv \{0, 1\}$$

$$g_2(X, Y) \in \{0, 0, 0, 1\}$$

$$(1-p)(1-q) \quad q(1-p) + p(1-q) + pq = q - pq + p - pq + pq$$

$$= p + q - pq = d$$

$$\{0, 1\}$$

$$= 1 - p - q + pq = 1 - d$$

$$p \log_2 \frac{1}{p} + (1-p) \log_2 \frac{1}{1-p} = -p \log_2 p - (1-p) \log_2 (1-p)$$

$$d \log_2 \frac{1}{d} + (1-d) \log_2 \frac{1}{1-d}$$

$$\{0, 1\}$$

$$pq = \beta$$

$$\beta \log_2 \frac{1}{\beta} + (1-\beta) \log_2 \frac{1}{1-\beta}$$

$X \backslash Y$	0	1
0	A	B
1	C	D

$$(1-p)(1-q) + q(1-p) + p(1-q)$$

$$= 1 - pq - pq - pq + q - pq + p - pq = 1 - pq = 1 - \beta$$

$$\{0, 1\}$$

$$\min(X, Y) \in \{1-D, D\}$$

$$A+B+C=1-D$$

$$\max(X, Y) \in \{A, B+C+D=1-A\}$$

$$D \log_2 \frac{1}{D} + (1-D) \log_2 \frac{1}{1-D}$$

$$A = 0.3 + a^2$$

$$B = 0.4$$

$$C = 0.2$$

$$D = a^2$$

$$\begin{cases} A+B+C+D=1 \\ 0 \leq \frac{A}{B} \leq 1 \\ \frac{C}{D} \end{cases}$$

$$Z = \min(X, Y)$$

$$E[Z^2] = 1 \cdot D + 0 \cdot (1-D) = D$$

$$Z = \min(X, Y)$$

$$= 0 \times A + 0 \times B + 0 \times C + 1 \times D = D$$

$g(0,0)$ $g(0,1)$ $g(1,0)$ $g(1,1)$

$$X, Y \stackrel{\text{indep}}{\sim} U(0,1)$$

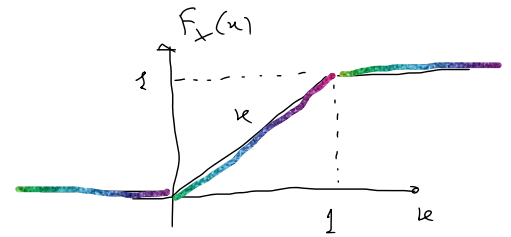
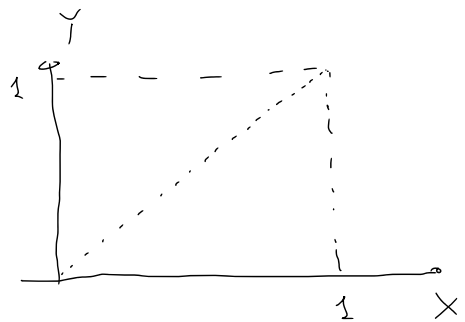
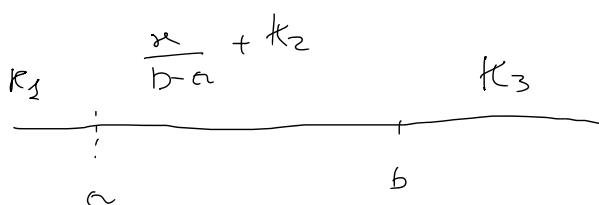
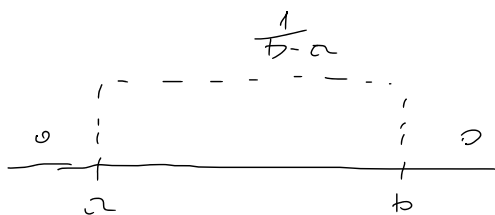
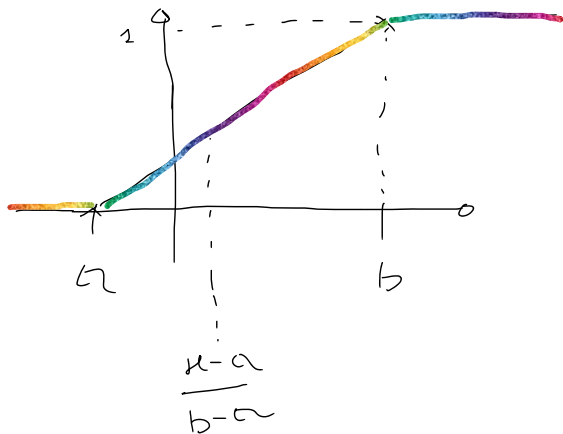
$$Z = \max(X, Y)$$

$$F_Z(z) = P[Z \leq z] = P[\max(X, Y) \leq z]$$

$$\max(X, Y) \leq z \Leftrightarrow \{X \leq z\} \wedge \{Y \leq z\}$$

$$= P[X \leq z] P[Y \leq z] = F_X(z) F_Y(z)$$

$$F_X(z) = F_Y(z) = F_X^2(z)$$



$$x \in (0, 1) \quad X \leq x \Leftrightarrow X \in (0, x]$$

$$\frac{1}{(b-a)} \int_a^x 1 \cdot dt = x - a$$

$$x \leq a \quad P[X \leq x] = 0$$

$$x \geq b \quad P[X \leq x] = 1$$

$$x \in (a, b) \quad P[X \leq x] =$$

$$= \int_{-\infty}^x f_X(t) dt = \int_a^x f_X(t) dt$$

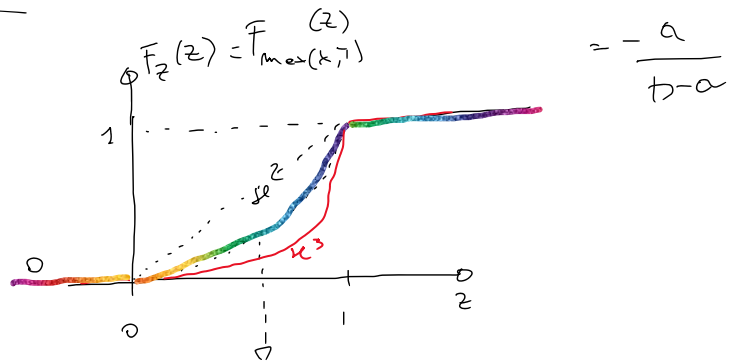
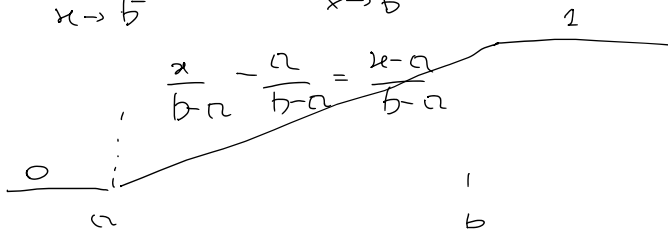
$$= \int_a^x \frac{1}{b-a} dt = \left[\frac{t}{b-a} \right]_{t=a}^{t=x} = \frac{x-a}{b-a}$$

$$\lim_{x \rightarrow -\infty} F_X(x) = 0 \Leftrightarrow k_1 = 0$$

$$\lim_{x \rightarrow +\infty} F_X(x) = 1 \Leftrightarrow k_3 = 1$$

$$\lim_{x \rightarrow a^-} F_X(x) = \lim_{x \rightarrow a^+} F_X(x) \Leftrightarrow \kappa_1 = \frac{a}{b-a} + \kappa_2 \quad \kappa_2 = \frac{-a}{b-a}$$

$$\lim_{x \rightarrow b^-} F_X(x) = \lim_{x \rightarrow b^+} F_X(x) \Leftrightarrow \frac{b}{b-a} + \kappa_2 = \kappa_3 \quad \kappa_2 = 1 - \frac{b}{b-a} = \frac{b-a-b}{b-a} = -\frac{a}{b-a}$$



X, Y, Z independent uniform $U(0,1)$

$$W = \max(X, Y, Z)$$

$$\begin{aligned} F_W(w) &= P[W \leq w] = P[\max(X, Y, Z) \leq w] = P[(X \leq w) \wedge (Y \leq w) \wedge (Z \leq w)] \\ &= P[X \leq w] P[Y \leq w] P[Z \leq w] = F_X(w) F_Y(w) F_Z(w) \\ &= F_X^3(w) \end{aligned}$$

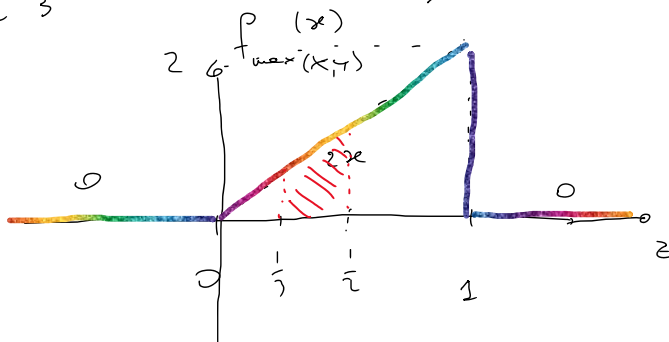
$$F_X(w) = F_Y(w) = F_Z(w)$$

$$\max(X, Y, \max(X, Y)) = \max(X, Y)$$

$$P\left[\frac{1}{3} \leq \max(X, Y) \leq \frac{1}{2}\right] = F_Z\left(\frac{1}{2}\right) - F_Z\left(\frac{1}{3}\right) = \left(\frac{1}{2}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{1}{4} - \frac{1}{9} = \frac{5}{36}$$

$$P[X \in (a, b)] = F_X(b) - F_X(a)$$

$$P\left[\frac{1}{3} \leq \max(X, Y, Z) \leq \frac{1}{2}\right] = F_W\left(\frac{1}{2}\right) - F_W\left(\frac{1}{3}\right) = \left(\frac{1}{2}\right)^3 - \left(\frac{1}{3}\right)^3 = \frac{1}{8} - \frac{1}{27} = \frac{27-8}{216} = \frac{19}{216}$$



$$\int_{\frac{1}{3}}^{\frac{1}{2}} 2x \, dx = \left[x^2 \right]_{\frac{1}{3}}^{\frac{1}{2}} = \left(\frac{1}{2}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{1}{4} - \frac{1}{9} = \frac{5}{36}$$

$$P[X \in (a, b)] = \int_a^b f_X(x) \, dx$$

$$\begin{cases} Z = g(X, Y) \\ W = X \end{cases}$$

$$(X, Y) \longrightarrow (Z, W)$$

$$\begin{cases} z = g(x, y) \\ w = x \end{cases} \quad \begin{cases} x = w \\ y = g^{-1}(z, w) \end{cases}$$

$$\int_{z, w} f(z, w)$$

$$\int f_{z, w}(z, w) dw = f_z(z)$$

$$Z = \min(X, Y)$$

$X, Y \sim U(0, 1)$ ed indip

$$F_Z(z) = P[Z \leq z] = P[\min(X, Y) \leq z] = P[(X \leq z) \text{ OR } (Y \leq z)] = P[X \leq z] + P[Y \leq z]$$

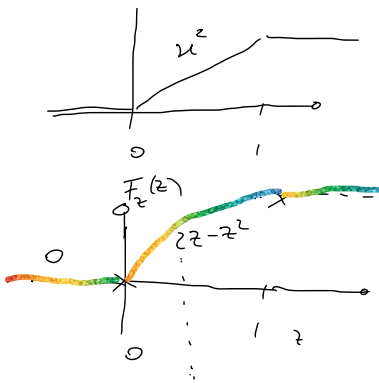
$$\min(X, Y) \leq z \Leftrightarrow (X \leq z) \text{ OR } (Y \leq z)$$

$$\Leftrightarrow$$

$$= P[X \leq z] + P[Y \leq z] - P[X \leq z \text{ AND } Y \leq z]$$

$$= F_X(z) + F_Y(z) - F_X(z) F_Y(z)$$

$$= 2 F_X(z) - F_X^2(z)$$



$$2x - x^2 = x(2 - x)$$

$$\downarrow$$

$$2 - 2x = 0 \quad x = 1$$

$$\min(X, Y) \leq z \Leftrightarrow (X \leq z) \text{ OR } (Y \leq z) \Leftrightarrow (X \leq z) \text{ AND } (Y \leq z)$$

$$P[\min(X, Y) \leq z] = 1 - P[\min(X, Y) > z] = 1 - F_Z(z)$$

$$P[X \leq z] = 1 - P[X > z] = 1 - F_X(z)$$

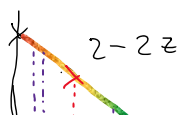
$$z = \frac{1}{2} \quad F_Z\left(\frac{1}{2}\right) = 2x - x^2 \Big|_{x=\frac{1}{2}} = 2\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)^2 = 1 - \frac{1}{4} = \frac{3}{4}$$

$$P[\min(X, Y) \leq z] = P[(X \leq z) \text{ AND } (Y \leq z)] = P[X \leq z] \cdot P[Y \leq z]$$

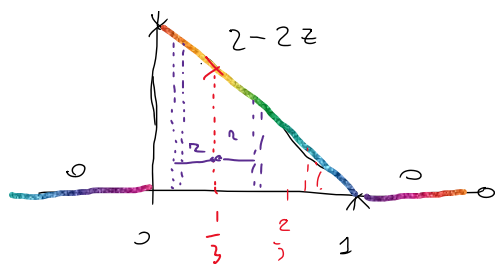
$$1 - F_Z(z) = (1 - F_X(z)) (1 - F_Y(z))$$

$$\Leftrightarrow F_Z(z) = F_X(z) + F_Y(z) - F_X(z) F_Y(z)$$

$$1 - F_X(z) - F_Y(z) + F_X(z) F_Y(z)$$



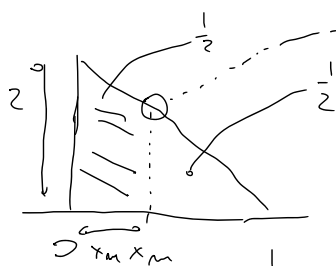
$$\int_{f_X(x)} dx = \frac{1}{2}$$



$$\int_{-\infty}^{x_m} f_X(x) dx = \frac{1}{2}$$

$$E[Z] = \int_{-\infty}^{+\infty} z f_Z(z) dz = \int_0^1 z f_Z(z) dz = \int_0^1 z (2-2z) dz = \left[z^2 - \frac{2}{3} z^3 \right]_{z=0}^1 = \frac{1}{3}$$

$$0 < x_m < 1$$



$$2 - 2x_m = 2(1-x_m)$$

$$= 1 - \frac{2}{3} = \frac{1}{3}$$

$$x_m \left[\frac{2 + 2(1-x_m)}{2} \right] = \frac{1}{2} \Leftrightarrow x_m [2 - x_m] = \frac{1}{2}$$

$$-x_m^2 + 2x_m = \frac{1}{2}$$

$$\frac{2(1-x_m)^2}{2} = \frac{1}{2} \quad (1-x_m)^2 = \frac{1}{2}$$

$$1 + x_m^2 - 2x_m = \frac{1}{2} \Leftrightarrow -x_m^2 + 2x_m = \frac{1}{2}$$

$$1 - x_m = \pm \frac{1}{\sqrt{2}}$$

$$x_m = 1 - \frac{1}{\sqrt{2}} = 1 - \frac{1}{\sqrt{2}}$$



$$VAR[Z] = E[Z^2] - \{E[Z]\}^2 = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{3}{18} - \frac{2}{18} = \frac{1}{18} \quad x_m \in (0, 1)$$

$$E[Z^2] = \int_{-\infty}^{+\infty} z^2 f_Z(z) dz = \int_0^1 z^2 f_Z(z) dz = \int_0^1 z^2 (2-2z) dz = \left[\frac{2z^3}{3} - \frac{2}{4} z^4 \right]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{4-3}{6} = \frac{1}{6}$$

$$E[XZ] = E\left[X \min(X, Y) \right] = \iint g(x, y) f_{X,Y}(x, y) dx dy$$

$$= \iint g(x, y) f_X(x) f_Y(y) dx dy$$

$$\left[\int \dots \right] \left[\int \dots \right]$$

$$= \iint x \min(x, y) f_X(x) f_Y(y) dx dy$$

$$= \int_0^1 \int_0^1 x \min(x, y) dx dy$$

$$= \int_0^1 dy \left[\int_0^1 x \min(x, y) dx \right] dy$$

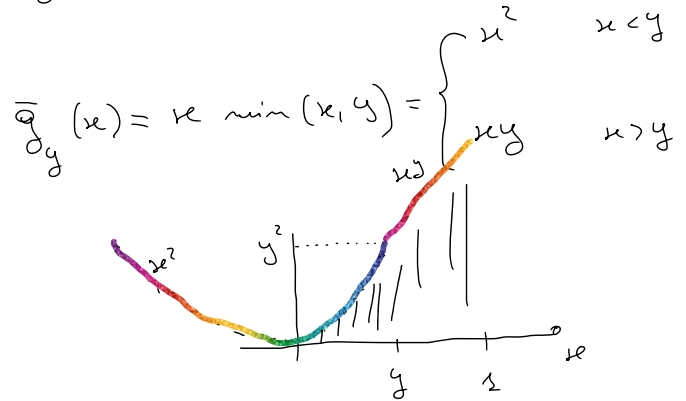
$$= \int_0^1 dy \left[\int_0^1 x \min(x, y) dx \right] dy$$

$$Y=y$$

$$W_{|Y=y} \quad E[W] = \int_{-\infty}^{+\infty} f_Y(y) E[W|Y=y] dy = \int_0^1 E[W|Y=y] dy$$

$$W = X \min(X, Y)$$

$$W_{|Y=y} = X \min(X, y) = \bar{g}_y(X)$$



$$E[W|Y=y] = E[\bar{g}_y(X)] = \int_{-\infty}^{+\infty} \bar{g}_y(x) f_X(x) dx$$

$$= \int_0^1 \bar{g}_y(x) dx = \int_0^y x^2 dx + \int_y^1 xy dx$$

$$= \int_0^1 x \min(x, y) dx = \left[\frac{x^3}{3} \right]_0^y + y \left[\frac{x^2}{2} \right]_y^1 =$$

$$= \frac{y^3}{3} + y \left[\frac{1}{2} - \frac{y^2}{2} \right]$$

$$= y^3 \left(\frac{1}{3} - \frac{1}{2} \right) + \frac{y}{2}$$

$$= -\frac{1}{6} y^3 + \frac{y}{2}$$

$$E[XZ] = E[W] = \int_{-\infty}^{+\infty} f_Y(y) E[W|Y=y] dy = \int_0^1 E[W|Y=y] dy$$

$$= \int_0^1 \left(\frac{y}{2} - \frac{y^3}{6} \right) dy = \left[\frac{y^2}{4} - \frac{y^4}{24} \right]_0^1 = \frac{1}{4} - \frac{1}{24} = \frac{5}{24}$$

$$\rho_{XZ} = \frac{E[XZ] - E[X]E[Z]}{\sqrt{\text{VAR}[X] \text{VAR}[Z]}}$$